

Jorge Armando Bravo

2026-1_E1IAP_SISTEMAS DE INFORMACION IND SOST

Actividad 6 - Actividad 6. Visualización de Datos Power BI

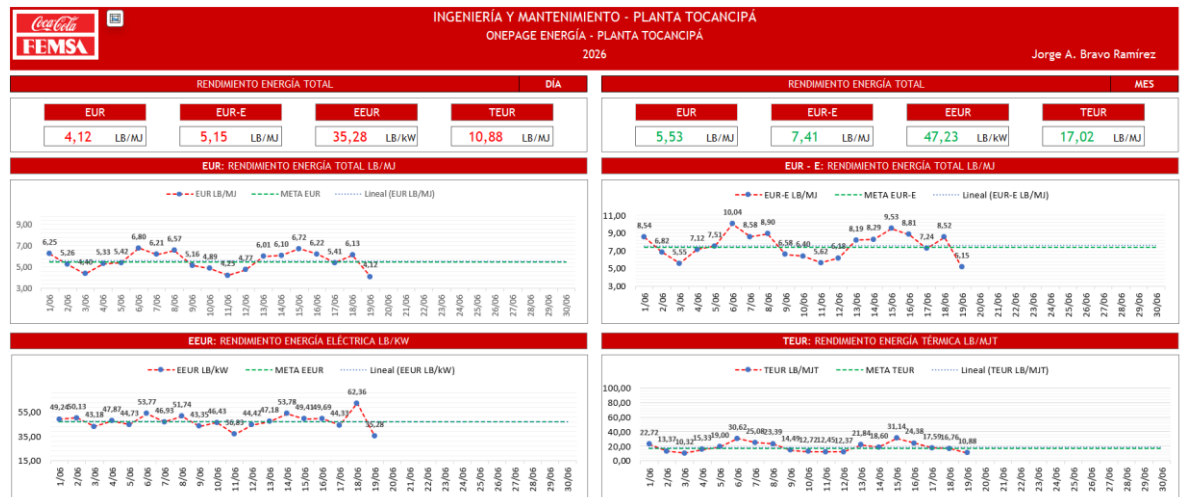
Se contaba con una base de datos en Excel donde se realiza seguimiento del comportamiento de los indicadores de energía eléctrica, para dar una solución dinámica, donde se permita la consulta de datos pasados, cambios en la visualización en rangos de tiempo, consulta de un solo actor de consumo eléctrico se genera proyecto en PowerBI brindando una solución de consulta.

Antes con Excel:

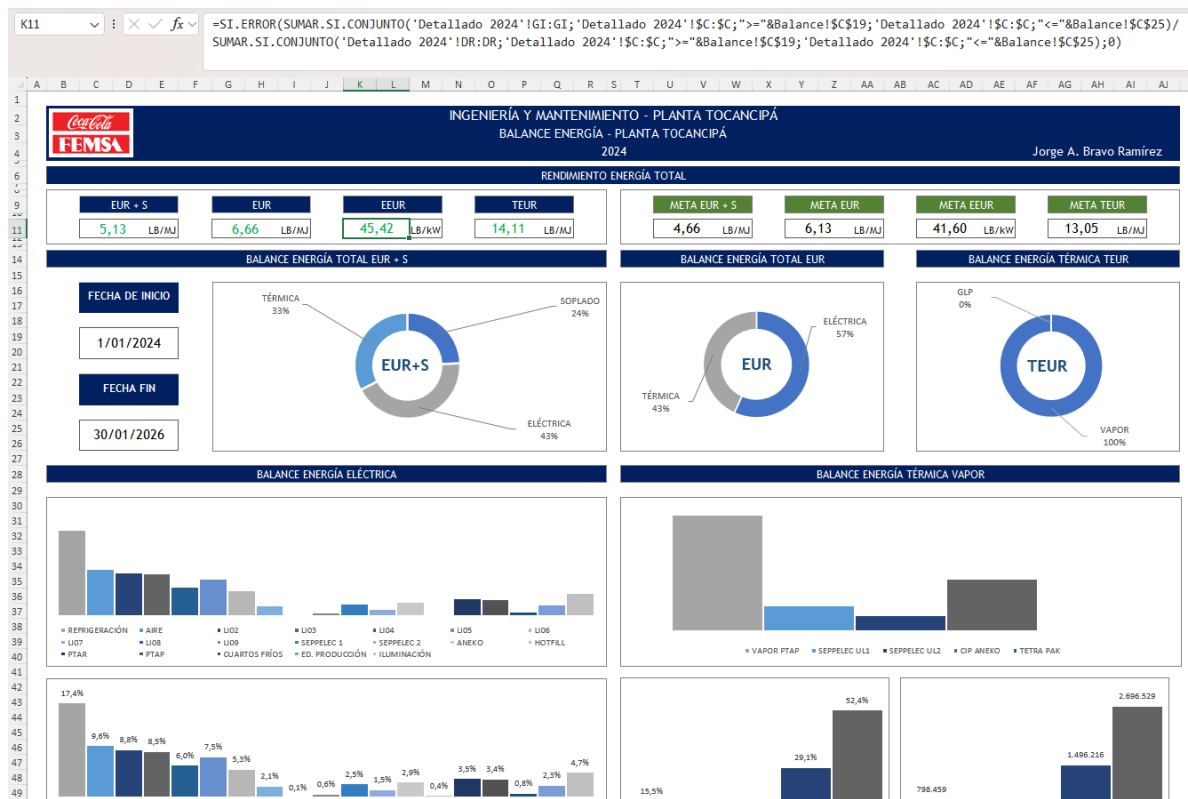
1. Base de datos en Excel.

FRONTIERA COMERCIAL													GENERACIÓN										ENERGÍA COMSUMIDA	
FECHA		CODENSA		EPM		EXPORTACIÓN		IMPORTACIÓN (CODENSA)		AIR LIQUIDE		ALCO - EXPORTACIÓN		ENERGÍA CONSUMIDA PEP		SUMAT (KWh)								
WEE	FECHA	KWh	KVAR	KWh	KVAR	KWh	KVAR	Factor	WWh	KWh	KVAR	Factor	WWh	KWh	KVAR	Factor	L00							
156	24/05/2026	7.902.71	1.636.64	7.900.49	1.637.08	1.365.69	834.99	10.036.125.61	0.00	18.287.80	4.084.14	174.565.318.69	66.083.15	61.842.095.28	25.472.651.41	61.842.095.28	118.605.26	184.688.41	175.9					
157	25/05/2026	7.908.67	1.637.82	7.906.45	1.638.26	1.365.69	834.99	10.036.125.61	0.00	18.293.76	4.085.34	174.622.161.85	56.843.15	61.962.682.23	25.524.866.64	61.962.682.23	120.586.95	177.430.10	166.8					
158	26/05/2026	7.914.83	1.639.07	7.912.61	1.639.51	1.365.69	834.99	10.036.125.61	0.00	18.299.92	4.086.59	174.680.990.45	58.828.61	62.082.442.73	25.578.937.21	62.082.442.73	119.760.50	178.589.13	168.9					
159	27/05/2026	7.920.15	1.640.37	7.918.31	1.640.89	1.366.47	835.33	10.043.571.06	0.00	18.306.29	4.087.86	174.741.728.15	60.737.70	62.203.348.37	25.634.845.35	62.203.348.37	120.905.64	181.443.34	170.1					
160	28/05/2026	7.927.76	1.641.86	7.925.93	1.642.38	1.366.47	835.33	10.043.571.06	0.00	18.313.89	4.089.35	174.814.321.50	72.593.15	62.323.148.14	25.691.626.25	62.323.148.14	119.799.77	192.392.92	182.1					
161	29/05/2026	7.934.61	1.643.12	7.932.77	1.643.64	1.366.47	835.33	10.043.571.06	0.00	18.320.72	4.090.62	174.879.554.91	65.235.61	62.442.231.59	25.747.492.47	62.442.231.59	119.083.45	184.317.06	175.3					
162	30/05/2026	7.938.88	1.643.69	7.937.03	1.644.20	1.366.47	835.34	10.043.571.06	0.00	18.324.99	4.091.19	174.920.315.98	40.739.07	62.562.666.71	25.797.152.81	62.562.666.71	120.435.12	181.194.19	151.5					
163	31/05/2026	7.944.55	1.644.51	7.942.70	1.645.03	1.366.47	835.35	10.043.571.06	0.00	18.330.66	4.092.01	174.974.431.59	54.103.61	62.683.733.87	25.856.066.60	62.683.733.87	121.067.16	175.170.77	163.5					
164	1/06/2026	7.950.56	1.645.49	7.948.72	1.646.01	1.366.47	835.35	10.043.571.06	0.00	18.336.67	4.093.00	175.031.785.74	57.368.15	62.803.760.99	25.911.831.53	62.803.760.99	120.027.12	177.395.27	167.0					
165	2/06/2026	7.956.48	1.646.56	7.954.63	1.647.08	1.366.47	835.35	10.043.571.06	0.00	18.342.58	4.094.08	175.088.161.17	56.375.45	62.923.838.40	25.966.421.96	62.923.838.40	120.077.41	176.452.84	168.5					
166	3/06/2026	7.962.23	1.647.68	7.960.41	1.648.20	1.366.47	835.35	10.043.571.06	0.00	18.348.56	4.095.20	175.143.305.23	55.144.06	63.043.874.37	26.018.543.27	63.043.874.37	120.055.97	175.180.03	165.4					
167	4/06/2026	7.969.05	1.649.06	7.967.20	1.649.58	1.366.47	835.35	10.043.571.06	0.00	18.355.14	4.096.60	175.208.099.75	64.794.51	63.164.157.00	26.073.012.15	63.164.157.00	120.282.63	185.077.14	174.6					
168	5/06/2026	7.975.40	1.650.44	7.973.55	1.650.96	1.366.48	835.37	10.043.666.52	95.45	18.361.50	4.097.99	175.268.732.45	60.632.70	63.283.601.96	26.124.642.49	63.283.601.96	119.444.96	179.982.20	168.5					
169	6/06/2026	7.987.22	1.654.13	7.985.37	1.654.65	1.366.48	835.37	10.043.666.52	0.00	18.373.31	4.101.48	175.381.521.48	112.789.04	63.322.765.10	26.140.948.47	63.322.765.10	99.163.14	151.952.18	144.6					
170	7/06/2026	8.002.31	1.659.30	8.000.44	1.659.82	1.366.48	835.37	10.043.666.52	0.00	18.388.39	4.106.87	175.525.438.23	143.916.75	63.322.765.10	26.140.963.24	63.322.765.10	0.00	149.916.75	132.9					
171	8/06/2026	8.012.03	1.661.76	8.010.16	1.662.27	1.366.48	835.43	10.043.666.52	0.00	18.398.09	4.109.31	175.618.010.01	92.571.77	63.385.085.50	26.172.689.40	63.385.085.50	62.320.40	154.892.17	141.4					
172	9/06/2026	8.018.58	1.661.76	8.016.63	1.662.28	1.366.48	835.56	10.043.666.52	0.00	18.404.56	4.109.31	175.679.778.62	61.788.61	63.504.960.70	26.237.042.79	63.504.960.70	119.875.20	181.643.81	171.2					
173	10/06/2026	8.025.23	1.661.78	8.023.36	1.662.30	1.366.48	835.63	10.043.666.52	0.00	18.411.28	4.109.32	175.743.904.95	64.126.33	63.623.216.12	26.305.099.38	63.623.216.12	120.285.42	184.391.73	173.9					
174	11/06/2026	8.030.61	1.661.78	8.028.74	1.662.30	1.366.48	835.76	10.043.666.52	0.00	18.416.66	4.109.32	175.795.288.11	51.585.16	63.737.404.77	26.364.674.34	63.737.404.77	112.178.65	165.561.81	153.7					
175	12/06/2026	8.036.77	1.661.78	8.034.90	1.662.31	1.366.48	835.88	10.043.666.52	0.00	18.422.82	4.109.32	175.854.088.08	58.799.97	63.858.437.42	26.430.680.00	63.858.437.42	121.032.65	179.832.62	168.0					

2. Graficas Hechas en Excel – Estáticas



3. Uso de fórmulas extensas en Excel para poder obtener tendencias de datos históricos.



Ahora con PowerBI

1. Se toma como origen la misma base de datos.



2. Se transforman y se adaptan datos.

Dashboard Energía PIF (BM) Version USalle

Archivo Inicio Transformar Agregar columna Vista Herramientas Ayuda

Table.RenameColumns(*Filas filtradas*,{{("AVG LÍNEA #2 KWh TOTAL", "AVG LÍNEA #2 KWh TOTAL"), ("AVG LÍNEA #2 KWh / DÍA", "AVG LÍNEA #2 KWh / DÍA")}}

	FECHA	1.2 CODENSA KWh	1.2 CODENSA KVAR	1.2 EPM KWh	1.2 EPM KVAR	1.2 EXPORTACIÓN KWh
1	1/01/2026	7108,18	1507,72	7106,23		1508,14
2	2/01/2026	7112,66	1508	7110,71		1508,42
3	3/01/2026	7117,76	1508,34	7115,8		1508,75
4	4/01/2026	7121,45	1508,58	7119,49		1509
5	5/01/2026	7127,51	1509,46	7125,54		1509,88
6	6/01/2026	7134,65	1510,74	7132,69		1511,16
7	7/01/2026	7140,68	1511,51	7138,72		1511,92
8	8/01/2026	7145,58	1511,89	7143,61		1512,31
9	9/01/2026	7151,67	1512,84	7149,7		1513,26
10	10/01/2026	7156,71	1513,47	7154,74		1513,89
11	11/01/2026	7162,37	1514,27	7160,4		1514,68
12	12/01/2026	7167,92	1515,08	7165,95		1515,5
13	13/01/2026	7173,74	1515,87	7171,76		1516,29
14	14/01/2026	7178,04	1516,25	7176,07		1516,67
15	15/01/2026	7185,15	1517,74	7183,16		1518,16
16	16/01/2026	7192,84	1519,49	7190,86		1519,91
17	17/01/2026	7201,13	1521,55	7199,15		1521,97
18	18/01/2026	7208,7	1523,23	7206,71		1523,65
19	19/01/2026	7215,77	1524,59	7213,78		1525,01
20	20/01/2026	7222,49	1525,77	7220,5		1526,19
21	21/01/2026	7229,9	1527,25	7227,9		1527,67
22	22/01/2026	7236,3	1528,19	7234,3		1528,61
23	23/01/2026	7243,63	1529,64	7241,63		1530,07
24	24/01/2026	7250,74	1531,15	7248,74		1531,57
25	25/01/2026	7258,55	1532,86	7256,54		1533,29
26	26/01/2026	7267	1534,89	7264,99		1535,31
27	27/01/2026	7275,24	1536,91	7273,23		1537,33
28	28/01/2026	7283,42	1538,92	7281,41		1539,34

Consultas [2]

Detallado 2026 - Original

Detallado 2026 - Gráficas

PROPIEDADES

Nombre

Detallado 2026 - Original

Todas las propiedades

PASOS APLICADOS

Tipo cambiado3

Columnas con nombre ca...

Tipo cambiado4

Columnas con nombre ca...

Tipo cambiado5

Columnas con nombre ca...

Columnas quitadas6

Columnas con nombre ca...

Columnas quitadas7

Columnas con nombre ca...

Columnas quitadas8

Columnas con nombre ca...

Tipo cambiado6

Columnas con nombre ca...

Tipo cambiado7

Columnas quitadas9

Columnas con nombre ca...

Columnas quitadas10

Columnas con nombre ca...

Filas superiores quitadas

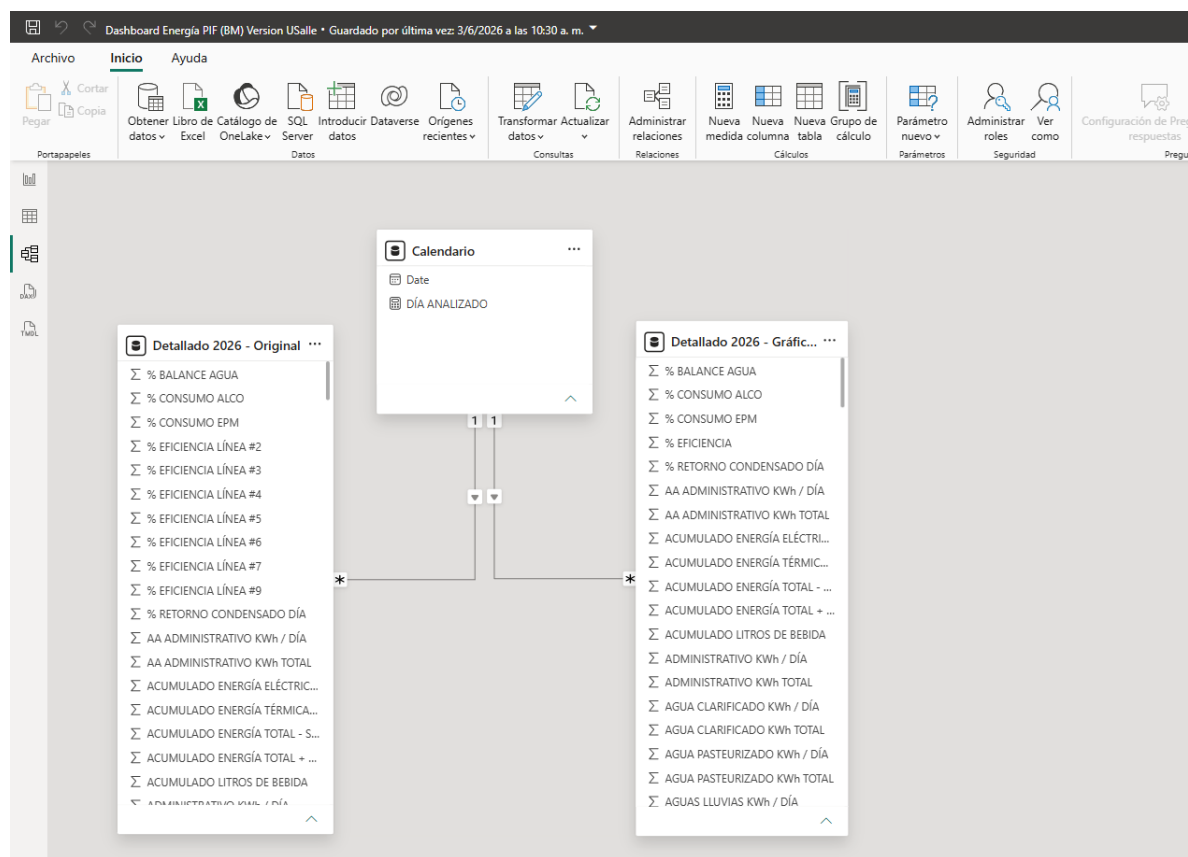
Filas inferiores quitadas

Tipo cambiado8

Filas filtradas

Columnas con nombre ca...

3. Se establecen relaciones de datos.



4. DashBoard





